

Force-Aware Insertion with Tactile Skins

Abstract

We show that tactile feedback at 200 Hz halves insertion failures compared with vision only.

1 Introduction

Insertion tasks fail mostly at contact onset, where vision cannot resolve millimetre offsets.

In our experiments the tactile policy reached 88 percent success versus 52 percent for vision.

The benefit holds across five connector types and three clearance levels in our study.